Brain Tumour Detection Using Convolutional Neural Networks

CEP: Artificial Intelligence and Machine Learning Lab

Muhammad Ahmad (FA23-BCE-113)
Uzair (FA23-BCE-098)

Computer Engineering Department
COMSATS University Islamabad – Lahore Campus

May 19, 2026

Outline

Problem Statement

- **Challenge:** Early cancer detection is crucial for patient survival
- Medical image analysis is time-consuming and requires expert radiologists
- Manual analysis is prone to human error and variability
- Limited access to specialised expertise in resource-constrained regions
- **Solution:** Automated image processing pipeline for objective feature extraction

Objective: Develop a robust, multi-stage image processing system for automated medical image analysis

Key Objectives

Primary Goals:

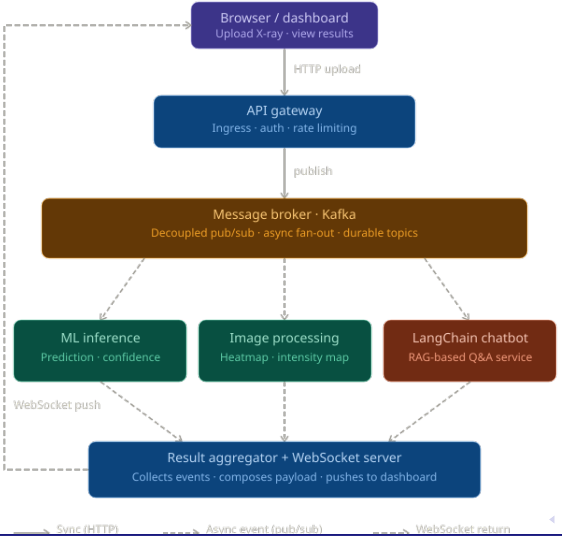
- Enhance diagnostic image contrast
- Extract structural features
- Identify tumour boundaries
- Reduce noise and artefacts
- Enable multi-aspect analysis

Technical Requirements:

- Scalable microservices architecture
- Real-time processing capability
- Cloud infrastructure integration
- Quantitative quality metrics
- Clinical validation support

Accuracy | **Speed** | **Equity** | **Scalability**

System Architecture Overview



Data Processing Flow

1. Upload Medical Image

↓

2. API Gateway Validates and Routes Request

↓

3. Image Processor applies 6-Stage Transformation

↓

4. Results Published to Apache Kafka Queue

↓

5. ML Service Consumes Features for Classification

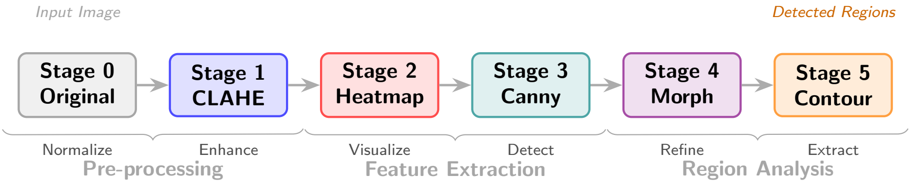
↓

6. Result Aggregator Persists to PostgreSQL

↓

7. Results Returned to User Dashboard

6-Stage Image Processing Pipeline



Stage 0: Original Image

Grayscale Normalisation:

- Load image as 8-bit grayscale (0–255)
- Normalise: $I_{norm} = \frac{I_{original}}{255}$
- Establish baseline for subsequent stages
- Compute intensity histogram

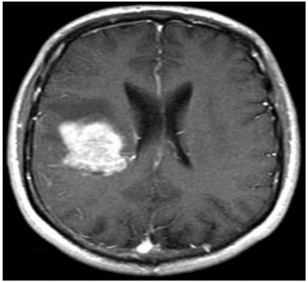
Key Information:

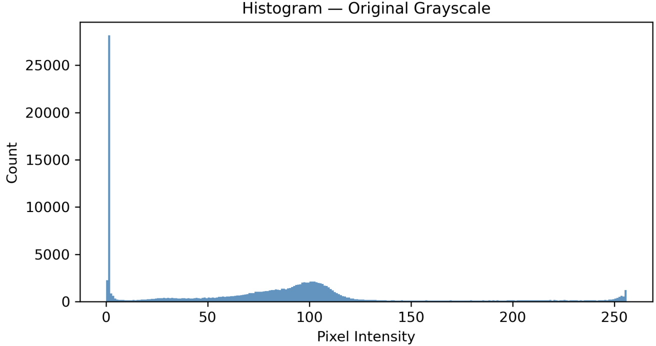
- Reference for PSNR and MSE
- Baseline for all comparisons
- Histogram shows distribution

Stage 0: Original Grayscale Image and Histogram

Stage 0 · Original

Original Grayscale Image





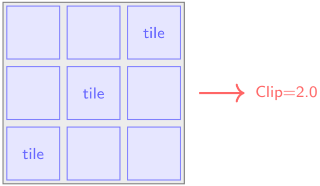
Stage 1: CLAHE Enhancement

Algorithm:

- Divide image into 8×8 tiles
- Per-tile histogram equalisation
- Clipping limit $= 2.0$
- Interpolate between tile boundaries

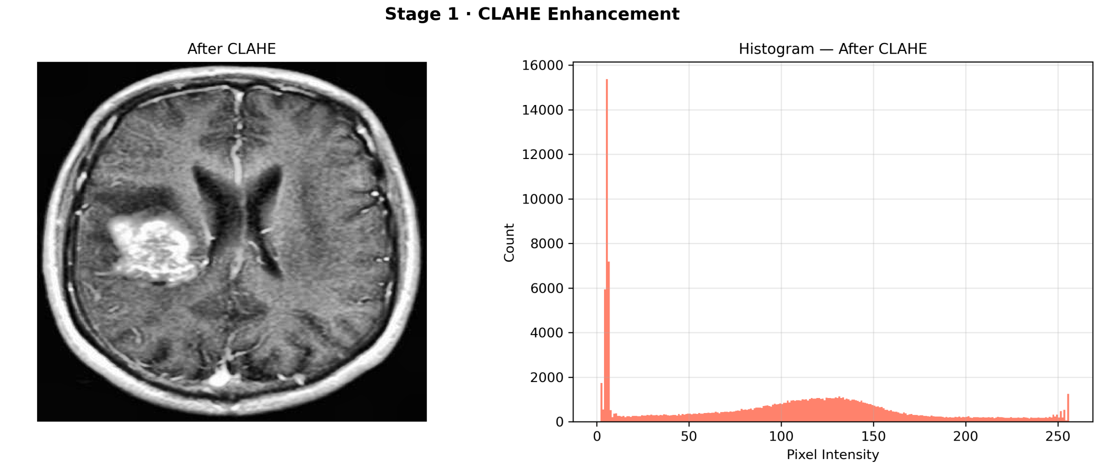
Benefits:

- Enhanced local contrast
- Prevents noise amplification
- Reveals subtle features
- Maintains natural appearance



8×8 Adaptive Tiles

Stage 1: CLAHE Enhancement Result



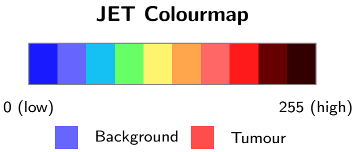
Stage 2: Heatmap Visualisation

JET Colour Mapping:

- Apply JET colourmap (blue to red)
- Dark colours: low intensity (background)
- Bright colours: high intensity (tumours)

Clinical Application:

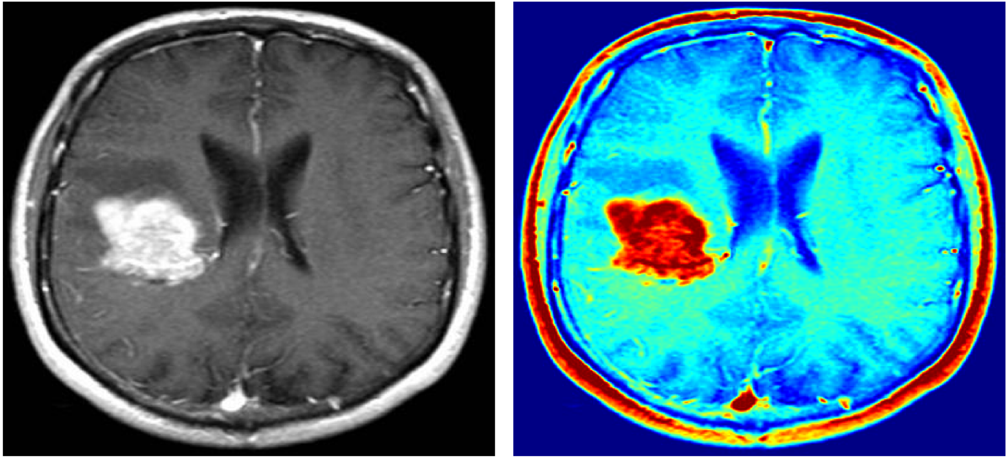
- Rapid hotspot identification
- Intuitive visual representation
- Radiologist interpretation aid



Stage 2: Heatmap Visualisation Result

Stage 2 · Heatmap Visualization — Intensity → Color Mapping

Original Grayscale | Heatmap (JET colormap)



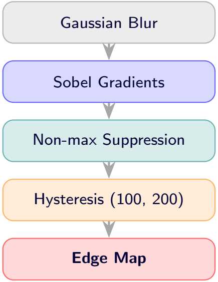
Stage 3: Canny Edge Detection

Algorithm Steps:

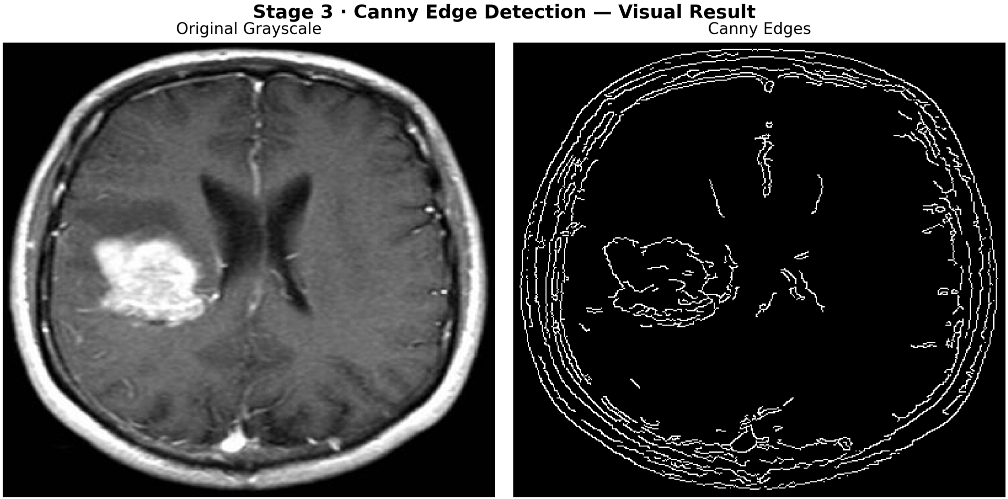
- Gaussian blur for noise reduction
- Sobel gradient in x and y directions
- Non-maximum suppression
- Hysteresis thresholding (100, 200)

Advantages:

- Precise structural boundaries
- Low false-positive rate
- Connected edge chains
- Robust to noise



Stage 3: Canny Edge Detection Result



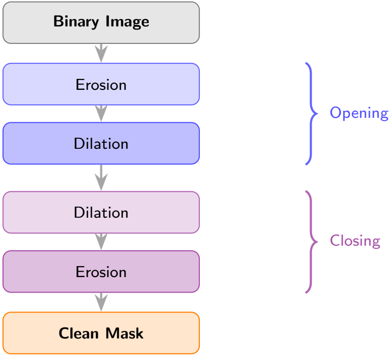
Stage 4: Morphological Operations

Filtering Sequence:

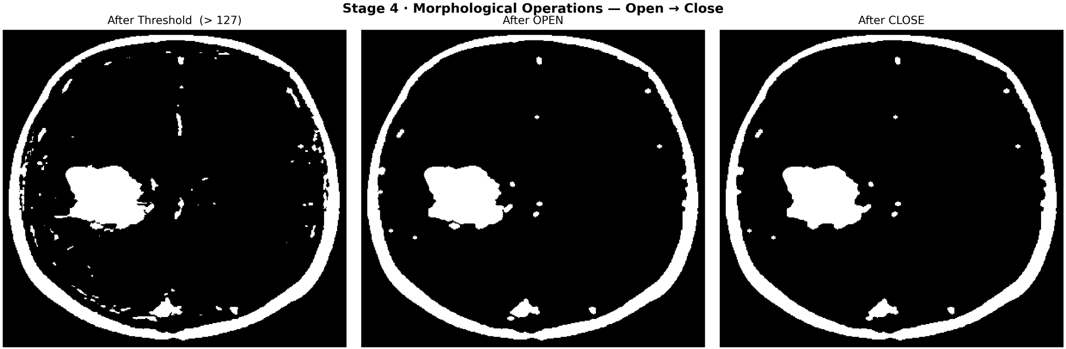
1. Binary threshold (= 127)
2. Opening: Erosion then Dilation
3. Closing: Dilation then Erosion
4. Kernel: 5×5 elliptical element

Effects:

- Removes small artefacts
- Fills holes in regions
- Preserves main structures



Stage 4: Morphological Operations Result

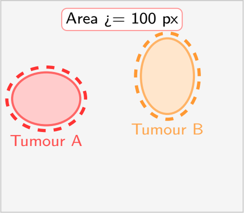


Connected Component Analysis:

- Find contours in binary image
- Filter by area (minimum 100 pixels)
- Extract boundary coordinates
- Compute shape properties

Output Features:

- Tumour regions identified
- Boundary coordinates extracted
- Area and perimeter computed
- Feature vectors for ML



Detected Contours

Stage 5: Contour Detection Result

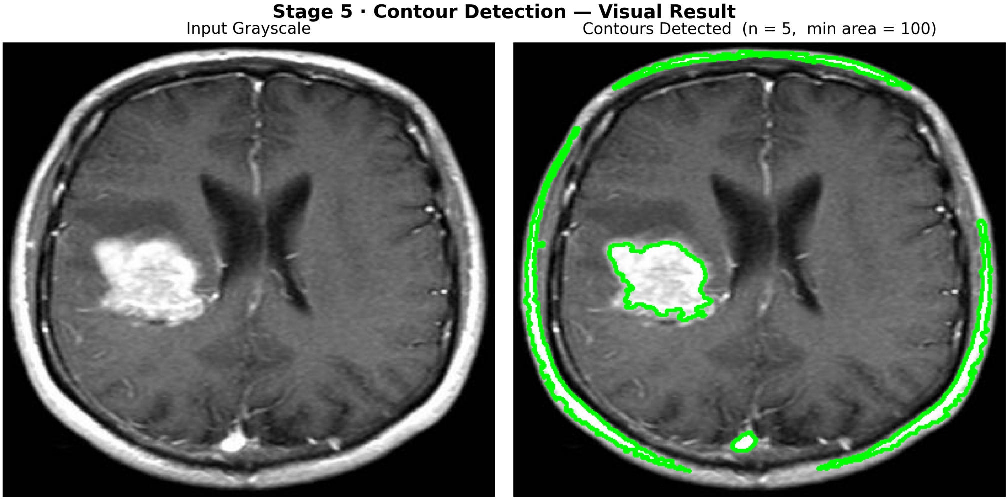
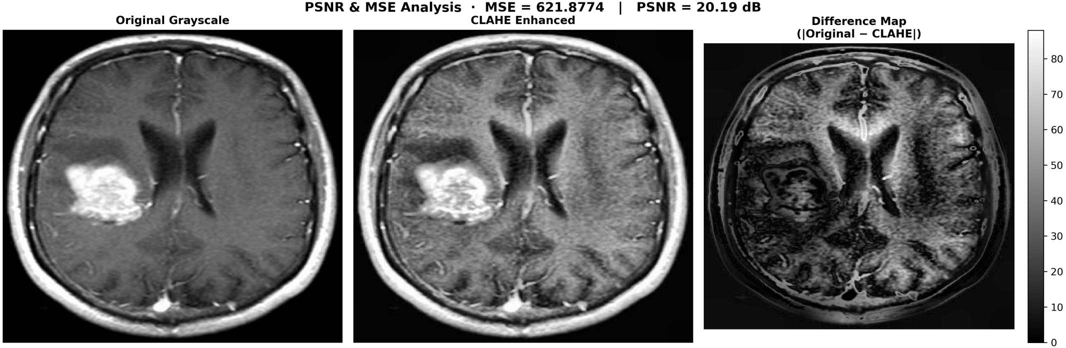


Image Quality Metrics Analysis



Quality Metrics – Interpretation

Measured Values:

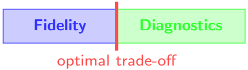
MSE621.88

PSNR20.19 dB

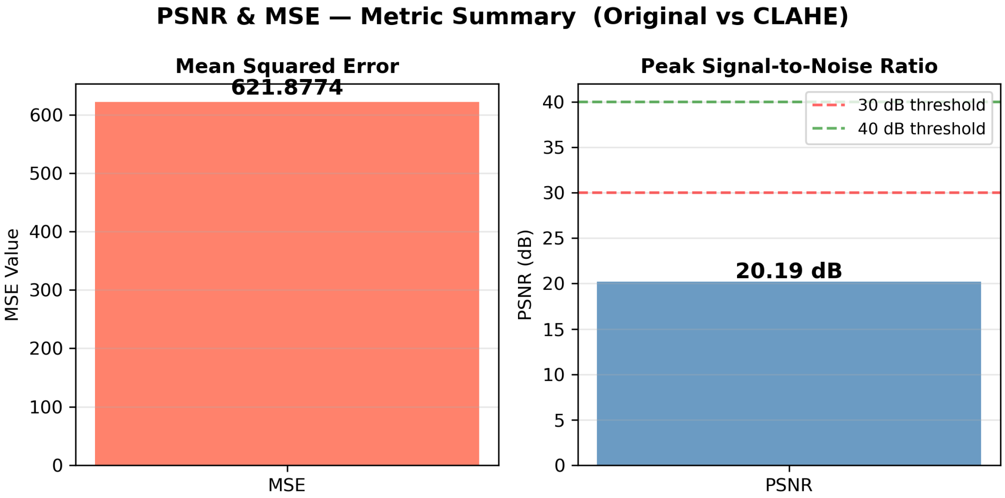
- Avg. pixel difference $\approx$ 25 units (0–255)
- **Expected** for aggressive feature extraction
- Reflects intentional structural modification

Clinical Context:

- Goal is feature clarity, not signal fidelity
- Lower PSNR indicates successful extraction
- Trade-off: preservation vs. diagnostics



PSNR / MSE Comparative Analysis



Quality Assessment Equations

Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{M \times N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} \left[I_{\text{orig}}(i,j) - I_{\text{proc}}(i,j) \right]^2$$

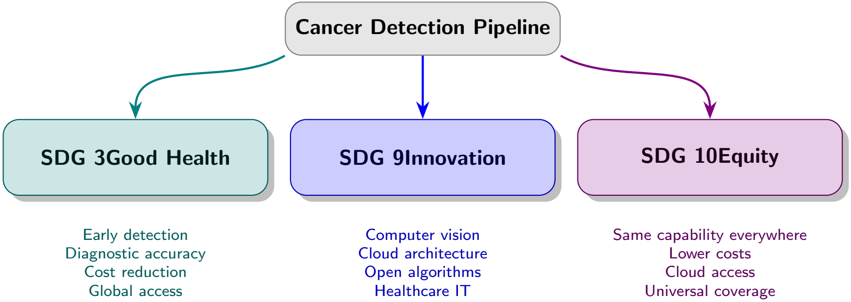
Peak Signal-to-Noise Ratio (PSNR):

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad \text{where MAX} = 255$$

> 30 dB: High Quality | 20–30 dB: Acceptable | < 20 dB: High Transform

Our result: 20.19 dB

SDG Alignment



Future Roadmap

Phase 1Short-term

CNN Integration
GPU Acceleration

Algorithmic Advances

Phase 2Mid-term

3D Volumetric
Spark Distributed

Infrastructure Scale

Phase 3Long-term

FDA Approval
HIPAA Compliance

Clinical Validation

Key Achievements

Technical Success:

- 6-stage pipeline fully implemented
- Multi-feature extraction working
- Quantitative metrics computed
- Cloud-native architecture deployed

Clinical Impact:

- Objective analysis capability
- Early detection support
- Accessible technology platform
- SDG 3, 9 and 10 alignment

Original → CLAHE → Heatmap → Canny → Morph → Contour

Project Vision

From Research to Global Impact

Research Prototype	→	**Production System**
Manual Analysis	→	**Automated Intelligence**
Expert Dependency	→	**Scalable Solution**
Regional Impact	→	**Global Healthcare**

Questions?

Questions and Discussion

Repository: `github.com/Muhammad-Ahmad17/cancer_cells`

Contact: `muhammadahmad171105@gmail.com`